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Extension of Modified Pseudo-Inverse Method with Generalized Linear Quadratic Stabilization

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Abstract— **When faults turn the ideal nominal operation to an unstable post-fault regime, the accommodation procedure must recover stability and as much performance of the reference model as possible. In this paper, the authors extend the Modified Pseudo-Inverse Method (MPIM) with the Generalized Linear Quadratic Regulator (GLQR) stabilization and prove its particular difference regarding the classical MPIM with simple LQR stabilization. Also, for the G/LQR stabilization, the authors propose the use of weighting matrices derived from the Robust Optimal Model Matching (ROMM) instead of arbitrary penalties as suggested in the original MPIM.**

I. Introduction

Fault Tolerant Control (FTC) refers to the approach by which the controlled system is able to exhibit desired properties, e.g., Model Matching / Following (MM/MF), both in normal operation and in the presence of faults [1]. Unfortunately, MM/MF stresses an idealistic objective in relation with the FTC design, because achieving exactly the same closed-loop performance as in the nominal case after the system impairment is almost impossible and approximate quality has to be accepted [2], [3].

In this paper, the authors extend the Modified Pseudo-Inverse Method (MPIM) [4] with the Generalized Linear Quadratic Regulator (GLQR) stabilization and prove its particular difference regarding the classical MPIM with simple LQR stabilization. Also, for the G/LQR stabilization, the authors propose the use of weighting matrices derived from the Robust Optimal Model Matching (ROMM) [5] instead of arbitrary penalties as suggested in the original MPIM.

It is organized as follows: Section II presents the approximate model matching problem with the Modified Pseudo-Inverse Method, while Section III addresses the optimal model matching problem with the Robust Optimal Model Matching method. The distinctive feature between the two control methods, namely the extension MPIM-GLQR and the classical MPIM-LQR, is highlighted in Section IV, while Section V concludes the paper.

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II. Approximate Model Matching

A. Nominal Operation

Consider the continuous linear time-invariant (LTI) deterministic system whose nominal operation is modeled by

$$M_n : \dot{x}(t) = A_n x(t) + B_n u_n^*(t), \text{ with } x(t_0 = 0) = x_0, \quad (1)$$

where $A_n \in \mathbb{R}^{n\times n}$ and $B_n \in \mathbb{R}^{n\times m}$ are the nominal system and control matrices, $x(t) \in \mathbb{R}^n$ and $u_n^*(t) \in \mathbb{R}^m$ are the state and control vectors, and $x_0 \in \mathbb{R}^n$ is the initial condition.

In nominal operation, the linear state-feedback control law

$$u_n^*(t) = -K_n^* x(t), \quad (2)$$

conducts to the closed-loop system

$$\hat{M}_n^* : \dot{x}(t) = \hat{A}_n^* x(t), \text{ with } \hat{A}_n^* = A_n - B_n K_n^*, \quad (3)$$

which is stable and provides the ideal dynamic performance.

B. Post-Fault Operation

Whether parametric faults are considered, assuming that an LTI model is still able to capture its behavior, the system to be controlled is described by the pair (A_f, B_f) provided by an on-line Fault Detection, Isolation, and Identification / Estimation (FDIE) module, that is, after the fault estimation time t_f, the post-fault system operation is modeled by

$$M_f : \dot{x}(t) = A_f x(t) + B_f u_f(t). \quad (4)$$

In post-fault operation, the Exact Model Matching (EMM) technique with State-Feedback (SF) aims at designing the linear control law

$$u_f(t) = -K_f x(t), \quad (5)$$

such that the resulting closed-loop behavior matches that of the Reference Model (RM) from (3).

The solution of the SF-EMM problem is obtained by solving the matrix equation

$$\hat{A}_f \triangleq A_f - B_f K_f = \hat{A}_n^*, \quad (6)$$

whose necessary and sufficient condition for a solution to exist is

$$\text{Im}\left(A_f - \hat{A}_n^*\right) \subseteq \text{Im}\left(B_f\right), \quad (7)$$

and the solution is given by

$$(K_f^{PIM} \triangleq) K_f = B_f^{\dagger}(A_f - \hat{A}_n^*), \quad (8)$$

where $(.)^{\dagger}$ stands for the pseudo-inverse matrix.

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C. Pseudo-Inverse Method

Regarding solution (8), condition (7) will obviously hold only for very particular faults and therefore no exact solution will exist in most fault cases; for this reason, approximate rather than exact solutions might be of interest.

When exact model matching is impossible $\left(\hat{A}_f \neq \hat{A}_n^*\right)$, an approximate control solution may be computed as

$$K_f^{\sharp} = \arg\min_{K_f} J_f(K_f)\,, \tag{9}$$

by minimizing the criterion (see [4])

$$J_f(K_f) = \|E_f\|_F^2\,, \tag{10}$$

with the impairment matrix

$$E_f = (A_f - B_f K_f) - \hat{A}_n^* \; (\Leftrightarrow \hat{A}_f = \hat{A}_n^* + E_f)\,, \tag{11}$$

where $\|.\|_F$ is the Frobenius matrix-norm.

Unfortunately, for certain fault cases, PIM encounters instability problems. (See also [6].)

D. Modified Pseudo-Inverse Method

The improved PIM, namely the Modified Pseudo-Inverse Method (MPIM), restricts the computation of K_f such that to guarantee the stability of the post-fault system while achieving as much of the closed-loop nominal performance as possible.

Its development is based on the fact that matrix $\hat{A}_f$ is stable if (see [7])

$$|c_{f_j}| < \frac{1}{\sigma_{\max}\left(\sum_{i=1}^{m} |\hat{X}_{f_i}|\right)}\,, \text{ for } j = 1,2,\ldots,m\,, \tag{12}$$

for $\hat{X}_f$ the solution to the continuous algebraic Lyapunov equation

$$\hat{A}_f^T \hat{X}_f + \hat{X}_f \hat{A}_f + 2I_n = 0\,, \tag{13}$$

and $\hat{X}_{f_e}$, $\hat{X}_{f_i}$ defined as

$$\hat{X}_{f_e} = \left[\begin{array}{cccc} \hat{X}_{f_1} & \hat{X}_{f_2} & \ldots & \hat{X}_{f_m} \end{array}\right], \text{ with}$$
$$\hat{X}_{f_i} = (E_{f_i}^T \hat{X}_f + \hat{X}_f E_{f_i})/2 = [\hat{X}_f E_{f_i}]_S\,, \text{ for } i = 1,2,\ldots,m;$$

it is assumed that matrix $E_f\,(= \hat{A}_f - \hat{A}_n^*)$ is bounded and may be expressed as $E_f = \sum_{i=1}^{m} c_{f_i} E_{f_i}$, where E_{f_i} are constant matrices and coefficients c_{f_i} vary in intervals around zero, that is $c_{f_i} \in [-\delta_f, \delta_f]$.

Precisely, choosing the stability bound δ_f to be, e.g., the right hand-side of (12), condition

$$|k_{f_{i,j}}^{MPIM}| \leq \tilde{\delta}_f\,, \text{ for } i = 1,2,\ldots,m, \text{ and } j = 1,2,\ldots,n\,, \tag{14}$$

guarantees the stability of $\hat{A}_f$, where $\tilde{\delta}_f = \delta_f - \varepsilon_f$, for some small ε_f; above, $[X]_S$ denotes the symmetric-part of the argument, i.e., $[X]_S = \left(X + X^T\right)/2$, $|X|$ indicates the matrix with its elements in absolute value, and $\sigma_{\max}$ stands for the maximum singular value.

Therefore, the control gain K_f^{MPIM} computed as

$$k_{f_{i,j}}^{MPIM} = k_{f_{i,j}}^{PIM}\,, \text{ if } |k_{f_{i,j}}^{PIM}| \leq \tilde{\delta}_f\,, \tag{15a}$$
$$k_{f_{i,j}}^{MPIM} = \operatorname{sgn}\left(k_{f_{i,j}}^{PIM}\right)\tilde{\delta}_f\,, \text{ otherwise}\,, \tag{15b}$$

solves the reconfigurable control problem with stability guarantees, where $\operatorname{sgn}(.)$ stands for the sign-function of the real-argument.

In what follows, for the extension of MPIM to be proposed in the paper, the robust optimal perspective regarding the model matching problem is also introduced briefly.

III. Robust Optimal Model Matching

Let $\hat{A}_n^*$ be the ideal nominal closed-loop matrix with $|\lambda(\hat{A}_n^*)| \geq \alpha_f$, where $|\lambda(\hat{A}_n^*)|$ is the maximum absolute value of the real-part of the eigenvalues of $\hat{A}_n^*$; the exponent α_f is defined as the desired minimal decay-rate of the post-fault closed-loop system, namely the exponential convergence-rate of system state to the equilibrium point [8].

Then, the post-fault state-feedback is chosen to minimize the performance index (see [5])

$$J_f(\alpha_f, \hat{A}_n^*) = \int_{t_f}^{\infty} e^{2\alpha_f t}\left(\|\dot{x}(t) - \hat{A}_n^* x(t)\|_2^2 + \|\hat{A}_n^* x(t)\|_2^2\right) dt\,, \tag{16}$$

and the optimal control $u_f(t) = -K_f x(t)$ which minimizes $J_f(\alpha_f, \hat{A}_n^*)$ subject to (A_f, B_f) is computed as

$$K_f = R_f^{-1}(B_f^T X_f + S_f)\,, \tag{17}$$

where the unique stabilizing symmetric semipositive-definite matrix X_f represents the solution to the Generalized Continuous Algebraic Riccati Equation (GCARE)

$$GCARE_f\,:\, \tilde{A}_f^T X_f + X_f \tilde{A}_f - X_f B_f R_f^{-1} B_f^T X_f + \tilde{Q}_f = 0\,, \tag{18}$$

with

$$\tilde{A}_f = A_f + \alpha_f I_n - B_f R_f^{-1} S_f\,,$$
$$\tilde{Q}_f = Q_f - S_f^T R_f^{-1} S_f\,, \tag{19}$$

where the required post-fault weighting matrices from (18) respectively (19) are identified as (see [9])

$$Q_f = A_f^T A_f - A_f^T \hat{A}_n^* - \hat{A}_n^{*T} A_f + 2\hat{A}_n^{*T} \hat{A}_n^*\,, \tag{20a}$$
$$R_f = B_f^T B_f\,, \tag{20b}$$
$$S_f = B_f^T (A_f - \hat{A}_n^*)\,, \tag{20c}$$

obtained after the reduction of the performance integral in (16) to a corresponding generalized-form.

IV. Modified Pseudo-Inverse Method with Generalized Linear Quadratic Stabilization

A. Control with LQR Stabilization and MPIM Correction

Going back to the model matching problem with the modified pseudo-inverse method, in the classical paper on MPIM [4], for certain instability cases, it was suggested to use first an efficient method to get the stabilizing gain, e.g., through solving the corresponding Continuous Algebraic Riccati Equation (CARE) associated with the post-fault pair (A_f, B_f), that is

$$CARE_f\,:\, A_f^T X_f + X_f A_f - X_f B_f R_f^{-1} B_f^T X_f + Q_f = 0\,, \tag{21}$$

where the solution to CARE_f is computed as

$$K_f^{CARE} = R_f^{-1} B_f^T X_f\,, \tag{22}$$

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and then to make a control correction as

$$K_f^{MPIM\text{-}LQR} = K_f^{CARE} + \tilde{\delta}_f \Delta K_f^{PIM\text{-}CARE}, \tag{23}$$

where the control accommodation difference $\Delta K_f^{PIM\text{-}CARE}$ is obtained as

$$\Delta K_f^{PIM\text{-}CARE} = B_f^{\dagger}\big((A_f - B_f K_f^{CARE}) - \hat{A}_n^*\big); \tag{24}$$

as no restrictions are indicated, see that the penalty matrices $Q_f^T = Q_f = \bar{Q}_f^T \bar{Q}_f \geq 0$ and $R_f^T = R_f > 0$ might be chosen as in (20a) and (20b).

Also, observe that (24) can be developed more as

$$\Delta K_f^{PIM\text{-}CARE} = B_f^{\dagger}(A_f - \hat{A}_n^*) - B_f^{\dagger} B_f K_f^{CARE}, \tag{25}$$

which, using (8) and the identity property of the pseudo-inverse, becomes

$$\Delta K_f^{PIM\text{-}CARE} = K_f^{PIM} - K_f^{CARE}. \tag{26}$$

Thus, after considering from (26) the correction of both terms, precisely PIM and CARE, the MPIM-LQR control (23) provides to have somehow a bi-objective form as

$$K_f^{MPIM\text{-}LQR} = \tilde{\delta}_f K_f^{PIM} + (1 - \tilde{\delta}_f) K_f^{CARE}. \tag{27}$$

B. Control with GLQR Stabilization and MPIM Correction

In this paper, the authors propose to obtain the initial stabilizing post-fault control matrix as K_f^{GCARE} from (17), solution of GCARE$_f$ similar to (18) associated with the faulty pair (A_f, B_f), after an appropriate choice of the penalty matrices Q_f, R_f, S_f as in (20).

In this respect, the extended control matrix becomes

$$K_f^{MPIM\text{-}GLQR} = K_f^{GCARE} + \tilde{\delta}_f \Delta K_f^{PIM\text{-}GCARE}, \tag{28}$$

where the control accommodation difference $\Delta K_f^{PIM\text{-}GCARE}$ is obtained as

$$\Delta K_f^{PIM\text{-}GCARE} = B_f^{\dagger}\big((A_f - B_f K_f^{GCARE}) - \hat{A}_n^*\big), \tag{29}$$

and thus, following the same logic as for the MPIM-LQR control, (28) gets the similar expression

$$K_f^{MPIM\text{-}GLQR} = \tilde{\delta}_f K_f^{PIM} + (1 - \tilde{\delta}_f) K_f^{GCARE}. \tag{30}$$

Furthermore, look with more consideration at the form of K_f^{GCARE}, which can be written as

$$K_f^{GCARE} = R_f^{-1}(B_f^T X_f + S_f), \tag{31}$$

but also, after replacing R_f and S_f from (20b) respectively (20c), as

$$K_f^{GCARE} = R_f^{-1} B_f^T X_f + B_f^{\dagger} B_f^{-T} B_f^T (A_f - \hat{A}_n^*), \tag{32}$$

which means that

$$K_f^{GCARE} = K_f^{PIM} + K_f^{CARE}. \tag{33}$$

Therefore, after replacing (33), the MPIM-GLQR control (30) becomes

$$K_f^{MPIM\text{-}GLQR} = K_f^{PIM} + (1 - \tilde{\delta}_f) K_f^{CARE}. \tag{34}$$

C. Interpretation

Look again at the similarity between the classical MPIM-LQR (27) and the extended MPIM-GLQR (34) controls and remark the clear difference between the two which is proved through the fact that in the MPIM-GLQR control the PIM component is no more penalized.

Moreover, see that if the stability-bound $\tilde{\delta}_f$ is much smaller than unity, $\tilde{\delta}_f \ll 1$, its influence becomes negligible, such that (27) respectively (34) get the approximate form

$$K_f^{MPIM\text{-}LQR} \backsimeq K_f^{CARE}, \tag{35}$$

$$K_f^{MPIM\text{-}GLQR} \backsimeq K_f^{PIM} + K_f^{CARE}. \tag{36}$$

However, if the fault is severe and the ideal performance denoted by $\hat{A}_n^*$ is too demanding, the control effort spent for stabilization will conduct to such K_f^{CARE} that might be several orders of magnitude bigger than K_f^{PIM} in norm, which will imply that the PIM correction becomes insignificant regarding the G/LQR stabilization, that is $K_f^{MPIM\text{-}GLQR} \approx K_f^{MPIM\text{-}LQR}$.

V. Conclusion

This paper interrogated the Modified Pseudo-Inverse Method (MPIM) used as an accommodation technique that recover as much performance as before the faults in the situation of impairments turning the ideal nominal operation to an unstable post-fault regime - unfortunately, stability is not always guaranteed.

The authors have extended the MPIM with the Generalized Linear Quadratic Regulator (GLQR) stabilization using the penalty matrices derived from the Robust Optimal Model Matching (ROMM) and have shown its distinctive feature regarding the classical MPIM used in conjuction with simple LQR.

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